

## SWIDWIN, Poland

WMO: 122120

Lat: 53.791N

Lon: 15.826E

Elev: 117

StdP: 99.92

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.0      | -10.2 | -16.5                         | 0.9 | -11.7 | -13.6 | 1.2 | -8.8 | 14.5                  | 6.5  | 12.9 | 5.0  | 2.5                   | 100  | 0.592 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.9                    | 29.1            | 19.4 | 27.0 | 18.6 | 25.3 | 17.8 | 20.7                | 26.9 | 19.7 | 25.2 | 18.7 | 23.3 | 4.3                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.6                            | 13.6 | 22.7 | 17.8 | 12.9 | 21.5 | 16.9 | 12.2 | 20.9 | 60.1          | 27.0 | 56.7 | 25.1 | 53.5 | 23.4 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.6          | 9.2  | 8.1 | DB                         | -15.8 | 32.3               | 4.0 | 2.1                                                | -18.7 | 33.8       | -21.1 | 35.0       | -23.3 | 36.2       | -26.2 | 37.7 | (4) |
| (5)               |      |     | WB                         | -15.6 | 22.5               | 4.1 | 1.2                                                | -18.6 | 23.3       | -21.0 | 24.0       | -23.3 | 24.7       | -26.3 | 25.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.6    | -1.4  | -0.2  | 3.2   | 8.6   | 12.6  | 16.0  | 18.2  | 17.9  | 13.7  | 8.8   | 4.4   | 1.2  |
|                                                                                  | DBStd       | 7.82   | 4.94  | 4.37  | 4.34  | 4.17  | 4.02  | 3.38  | 3.20  | 2.97  | 3.05  | 3.52  | 3.31  | 4.32 |
|                                                                                  | HDD10.0     | 1454   | 352   | 284   | 213   | 74    | 16    | 1     | 0     | 0     | 4     | 66    | 170   | 274  |
|                                                                                  | HDD18.3     | 3635   | 611   | 518   | 470   | 291   | 183   | 87    | 42    | 44    | 142   | 297   | 419   | 532  |
|                                                                                  | CDD10.0     | 960    | 0     | 0     | 2     | 33    | 98    | 181   | 256   | 246   | 116   | 28    | 1     | 0    |
|                                                                                  | CDD18.3     | 98     | 0     | 0     | 0     | 0     | 6     | 18    | 39    | 32    | 3     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 885    | 0     | 0     | 0     | 4     | 72    | 171   | 378   | 234   | 26    | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 212    | 0     | 0     | 0     | 0     | 11    | 37    | 108   | 54    | 2     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.8    | 4.4   | 4.1   | 4.0   | 3.8   | 3.5   | 3.5   | 3.4   | 3.3   | 3.5   | 3.8   | 4.1   | 4.4  |
| Precipitation                                                                    | PrecAvg     | 671    | 52    | 43    | 49    | 35    | 56    | 67    | 81    | 74    | 58    | 47    | 52    | 58   |
|                                                                                  | PrecMax     | 905    | 116   | 92    | 99    | 99    | 99    | 136   | 164   | 164   | 145   | 114   | 119   | 102  |
|                                                                                  | PrecMin     | 529    | 2     | 7     | 10    | 12    | 21    | 21    | 12    | 22    | 13    | 22    | 2     | 12   |
|                                                                                  | PrecStd     | 91     | 30    | 19    | 24    | 19    | 20    | 26    | 42    | 31    | 26    | 25    | 23    | 25   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 10.2  | 10.6  | 16.8  | 23.7  | 28.3  | 30.4  | 32.1  | 31.3  | 26.8  | 19.8  | 13.0  | 11.1 |
|                                                                                  |             | MCWB   | 8.8   | 7.8   | 10.0  | 14.7  | 18.0  | 20.8  | 20.4  | 20.7  | 19.0  | 14.9  | 11.1  | 9.8  |
|                                                                                  | 2%          | DB     | 7.9   | 8.5   | 14.0  | 21.0  | 25.4  | 27.3  | 29.6  | 28.0  | 23.3  | 17.7  | 11.1  | 9.2  |
|                                                                                  |             | MCWB   | 6.8   | 6.5   | 8.2   | 12.8  | 16.9  | 18.5  | 19.5  | 19.0  | 16.3  | 13.9  | 9.8   | 8.2  |
|                                                                                  | 5%          | DB     | 6.4   | 6.8   | 11.4  | 18.7  | 22.7  | 25.2  | 27.3  | 25.8  | 21.4  | 15.6  | 10.0  | 7.8  |
|                                                                                  |             | MCWB   | 5.5   | 5.1   | 7.6   | 11.4  | 15.2  | 17.5  | 18.7  | 18.0  | 15.5  | 12.8  | 8.9   | 6.7  |
|                                                                                  | 10%         | DB     | 5.0   | 5.5   | 9.4   | 16.3  | 20.5  | 22.9  | 25.3  | 24.0  | 19.2  | 13.9  | 9.0   | 6.4  |
|                                                                                  |             | MCWB   | 4.2   | 4.0   | 6.6   | 10.0  | 14.0  | 16.3  | 18.1  | 17.6  | 14.7  | 11.9  | 8.1   | 5.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.8   | 9.0   | 11.3  | 15.8  | 19.3  | 22.2  | 21.6  | 21.7  | 19.6  | 16.1  | 11.1  | 10.1 |
|                                                                                  |             | MCDB   | 10.2  | 9.8   | 15.1  | 22.9  | 25.7  | 28.9  | 28.1  | 28.8  | 25.5  | 18.5  | 12.2  | 10.9 |
|                                                                                  | 2%          | WB     | 6.9   | 6.8   | 9.5   | 13.5  | 17.6  | 19.9  | 20.6  | 20.5  | 17.4  | 14.4  | 9.9   | 8.4  |
|                                                                                  |             | MCDB   | 7.8   | 8.0   | 12.1  | 19.4  | 23.8  | 25.4  | 27.4  | 26.0  | 21.2  | 16.6  | 10.8  | 9.1  |
|                                                                                  | 5%          | WB     | 5.5   | 5.3   | 8.1   | 12.0  | 16.4  | 18.4  | 19.7  | 19.3  | 16.3  | 13.2  | 9.0   | 6.8  |
|                                                                                  |             | MCDB   | 6.2   | 6.6   | 10.7  | 17.4  | 21.4  | 23.4  | 25.7  | 23.7  | 19.6  | 15.2  | 9.8   | 7.5  |
|                                                                                  | 10%         | WB     | 4.3   | 4.1   | 6.7   | 10.6  | 14.8  | 17.1  | 18.7  | 18.4  | 15.4  | 12.1  | 8.1   | 5.6  |
|                                                                                  |             | MCDB   | 4.9   | 5.4   | 9.1   | 15.3  | 18.8  | 21.5  | 23.5  | 22.1  | 18.5  | 13.7  | 8.8   | 6.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.4   | 5.4   | 7.7   | 10.2  | 10.5  | 10.2  | 9.9   | 9.6   | 8.7   | 6.9   | 4.5   | 4.0  |
|                                                                                  |             | MCDBR  | 4.3   | 6.7   | 11.3  | 13.7  | 13.5  | 13.5  | 13.7  | 12.5  | 11.7  | 9.0   | 5.6   | 4.9  |
|                                                                                  |             | MCWBR  | 4.0   | 5.2   | 6.8   | 7.3   | 6.5   | 6.8   | 5.6   | 5.8   | 5.8   | 5.7   | 4.6   | 4.4  |
|                                                                                  | 5% WB       | MCDBR  | 4.3   | 6.0   | 9.9   | 12.7  | 12.2  | 12.1  | 12.0  | 10.7  | 9.9   | 8.2   | 5.3   | 4.8  |
|                                                                                  |             | MCWBR  | 4.1   | 5.0   | 6.6   | 7.1   | 6.3   | 6.7   | 5.7   | 5.7   | 5.6   | 5.7   | 4.5   | 4.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.307  | 0.333 | 0.364 | 0.403 | 0.401 | 0.397 | 0.420 | 0.417 | 0.377 | 0.358 | 0.336 | 0.305 |      |
|                                                                                  | taud        | 2.297  | 2.300 | 2.273 | 2.231 | 2.286 | 2.322 | 2.274 | 2.290 | 2.384 | 2.400 | 2.357 | 2.305 |      |
|                                                                                  | Ebn at Noon | 648    | 754   | 808   | 823   | 846   | 853   | 823   | 801   | 791   | 719   | 609   | 573   |      |
|                                                                                  | Edh at Noon | 59     | 82    | 104   | 123   | 123   | 120   | 124   | 115   | 93    | 74    | 57    | 49    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.54   | 1.21  | 2.43  | 4.16  | 5.24  | 5.53  | 5.11  | 4.32  | 3.13  | 1.62  | 0.65  | 0.38  |      |
|                                                                                  | RadStd      | 0.07   | 0.17  | 0.29  | 0.54  | 0.60  | 0.43  | 0.64  | 0.39  | 0.34  | 0.24  | 0.09  | 0.04  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (34 neighbors) | +0.39 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -141    | +62     | N/A     |

Nomenclature: See separate page